Convert mass and capacity

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ________________

1. Read the model: Convert 2.65 kg to grams and 3,750 ml to litres. Copy or complete the first step.

2. Write 845 mm in metres.

3. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Convert 2.65 kg to grams and 3,750 ml to litres. Copy or complete the first step.
   Answer 2,650 g and 3.75 litres.

2. Write 845 mm in metres.
   Answer 0.845 m.

3. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
   Answer About 48 km.

4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.
   Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.

Convert mass and capacity

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ______________

1. Convert 2.65 kg to grams and 3,750 ml to litres.

2. Write 845 mm in metres.

3. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

4. Solve this independently and show every stage: Convert 2.65 kg to grams and 3,750 ml to litres.

5. Use a second method or representation for: Write 845 mm in metres.

6. Check this result using an inverse, estimate or known fact: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

7. Prove or explain your reasoning: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

Teacher answers and checking prompts

1. Convert 2.65 kg to grams and 3,750 ml to litres.
   Answer 2,650 g and 3.75 litres.

2. Write 845 mm in metres.
   Answer 0.845 m.

3. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
   Answer About 48 km.

4. Solve this independently and show every stage: Convert 2.65 kg to grams and 3,750 ml to litres.
   Answer 2,650 g and 3.75 litres.

5. Use a second method or representation for: Write 845 mm in metres.
   Answer Expected result: 0.845 m. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
   Answer Expected result: About 48 km. A valid check must be shown.

7. Prove or explain your reasoning: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
   Answer 31 complete pieces, with 10 cm left. The explanation must show why the method works.

8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.
   Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Convert mass and capacity

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: ___________________

1. Prove or explain your reasoning: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

2. Prove the answer to this question, rather than only calculating it: Convert 2.65 kg to grams and 3,750 ml to litres.

3. Solve this in two different ways and compare them: Write 845 mm in metres.

4. Work backwards from the answer to reconstruct the question: About 48 km.

5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.

6. Create a new example that tests the same idea as: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
   Answer 31 complete pieces, with 10 cm left. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Convert 2.65 kg to grams and 3,750 ml to litres.
   Answer Expected result: 2,650 g and 3.75 litres. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Write 845 mm in metres.
   Answer Expected result: 0.845 m. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: About 48 km.
   Answer One valid reconstruction is: Use 5 miles is about 8 km. Estimate 30 miles in kilometres. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.
   Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.

6. Create a new example that tests the same idea as: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.